



Gaurav Santosh Gokhale

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Summary

C++ software engineer with 3+ years of experience developing performant and maintainable software for compute-intensive applications. Experienced with CUDA, Kokkos, MPI, OpenMP, and CMake, including GPU acceleration, distributed debugging, test automation and Linux development workflows. Currently based in Vienna, open to **relocate**.

WORK EXPERIENCE

DCS Computing GmbH

Jan. 2025–Present

HPC Software Developer, Aspherix Simulation Software

- Improved GPU performance by porting particle-mesh force calculation to GPU and accelerating broad-phase neighbor-list generation by 20x.
- Extended backend and build support for CUDA, HIP, and OpenMP, including AMD GPU enablement through Kokkos/HIP.
- Debugged distributed-memory simulation issues including segmentation faults, invalid memory accesses, and MPI calculation errors using Python–GDB integration and tpmi.
- Refactored meshing code for convex decomposition of concave geometries, integrated CoACD, and added unit tests with GoogleTest.

Altair Engineering GmbH

June 2022–June 2024

Working Student, nanoFluidX GPU Simulation Software Team

- Built C++ diagnostic tools for multi-GPU systems and Python-based automated test workflows.
- Implemented and maintained the Linux CMake build for a CUDA/C++ simulation codebase.
- Delivered multiphysics coupling between nanoFluidX and MotionSolve using Kratos/CoSimIO.

SELECTED PROJECTS

GPU accelerated Rayleigh Benard Convection Simulation

Sept. 2024–Nov. 2024

- Developed a C++/CUDA finite-difference solver for Rayleigh-Benard convection.
- Implemented Red/Black SOR for multithreaded stencil operations on a 2D grid.
- Added unit testing with GoogleTest and CI/CD on a self-hosted runner using GitHub Actions.

Performance-portable implementation of ALPACA (link)

Dec 2023–June 2024

- Ported an adaptive mesh refinement CFD code to a hardware-agnostic framework using Kokkos.
- Benchmarked single-phase simulations on NVIDIA GPUs and OpenMP CPUs.
- Achieved up to 3x speedup on selected cases on an NVIDIA A1000 GPU versus a 16-core CPU.

TECHNICAL SKILLS

Software C++20/17/14 | CUDA | Kokkos | MPI | OpenMP | CMake | GDB | Git | Docker | GTest

Languages English (Fluent) | German (Beginner) | Marathi (Native) | Hindi (Native)

EDUCATION

MSc. Computational Science and Engineering - Grade: 1.7

2021–2024

Technische Universität München

B.E. Mechanical Engineering - Grade: 1.4

2015–2019

Savitribai Phule Pune University